

Bridging the Gap between Education and Practice in Japan's CG Industry: A Focus on Diversity and Evolving Perceptions

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ABSTRACT

This study focuses on the competency gap between computer graphics (CG) educational institutions and CG companies in Japan, with an emphasis on the competencies required in the entertainment industry. The goal is to seek and examine viable solutions for establishing a sustainable CG education framework.

CCS CONCEPTS

• **Applied computing** Education; Interactive learning environments; Education; Collaborative learning;; • **Social and professional topics** Professional topics; Computing profession; Employment issues;

KEYWORDS

Education-Industry Collaboration, Skill Gap, Sustainable Education Program, Diversified Needs and Skills

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1 INTRODUCTION

The CG field in Japan is experiencing a gap between the ideal professional profile required in practice and the current state of CG education. This disparity has emerged due to the continuous development and diversification of the industry. Unfortunately, since the "National Survey on 3D CG Education" conducted in 2010 by CG-ARTS, 2011 and the Japan-US Comparative Study on 3D CG/Animation Education by Aoki et al., 2011, little academic research has been conducted in the field of CG education in Japan.

Against this background, this study aims to propose directions for educational reforms that contribute to the sustainable development of the CG industry. Additionally, it seeks to suggest improvements to curricula and educational frameworks that are both timely and relevant, while also illuminating the current state of Japan's CG education through international comparisons.

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To achieve these goals, the study addresses the following major research question: How can CG education in Japan be reformed to better align with industry needs and international standards?

2 METHODOLOGY

Surveys on the "Talent Required by Production" were ethically conducted targeting CG companies located in Tokyo in January and December of 2023. The study aimed to analyze how "technical hard skills," "art skills," and "soft skills expressed as human qualities" relate to the desired profile of talent. The desired human resource profile was broadly divided into three categories: "technical skills," "efficiency in CG Work," and "human qualities." The survey specifically focused on elements considered crucial within each category. Qualitative data collection was also carried out (Supplements Table 1).

Subsequently, the "National Survey on 3DCG Education" was conducted from October 2023 to January 2024, both domestically and internationally, under ethical considerations and via the Internet, similar to Aoki et al., 2011. The survey included questions on the size of the schools, the number of departments and faculty members, and the curriculum, allowing for comparisons with past records. Recruitment of participants in Japan was accomplished through school lists provided by CG-ARTS and cooperating companies, as well as at the venue of SIGGRAPH ASIA 2023 and through the ACM SIGGRAPH Education Committee's website. During the quantitative data analysis, differences between 2011 and 2023 were examined to identify significant changes.

For qualitative data, axial coding based on grounded theory was adopted. Finally, drawing from the insights gained, a detailed analysis of the gap between industry needs and educational competencies was conducted, leading to concrete suggestions for reducing this gap.

3 SURVEY RESULTS FOR COMPANIES AND EDUCATIONAL INSTITUTIONS

In the survey on "Talent Required by Production," companies with 1–30 employees accounted for 35%, 31–100 employees for 25%, and over 100 employees for 40%, reflecting the diversity in the size of Japanese CG companies. The "National Survey on 3DCG Education" in Japan received a total of 70 responses from 23 universities, including graduate schools and junior colleges, and 47 vocational schools, high schools, and private institutes.

Internationally, most responses came from universities and graduate schools (USA:10, Asia and Oceania:5, others:3). Given the significant difference in the sample size between the surveys in Japan and internationally, careful consideration was necessary to

ensure statistical validity and contextual appropriateness when conducting the comparative analyses.

4 TALENT REQUIRED BY PRODUCTION

In terms of "technical skills," the most demanded skills by production are "specialized technical skills (such as modelers and animators)" (93.9%). Interestingly, small companies tend to look for generalists, while large companies prefer specialists. However, this phenomenon is not necessarily unique to the CG industry. Moreover, some major companies prefer generalists, thus necessitating an understanding of the skill sets appropriate for the working environment. Furthermore, the demand for CG skills and professionals within the industry is diversifying and growing in complexity (Supplements Table 2).

As for "efficiency in CG work," the key factors are "ability to report and communicate" (84.6%) and "ability to meet deadlines" (81.5%), highlighting the importance of basic operational skills.

Regarding "human qualities," communication skills ranked the highest (92.3%), followed by "cooperativeness" (83.1%) and "initiative and action" (70.8%). In communication skills, oral presentations (73.2%) and other aspects were emphasized. However, compared to 3D CG practice, students show less interest and engagement in this area, underscoring the importance of programs that offer real-world communication internships and "realizations" about the necessity of communication.

These findings suggest that, in the CG field, both technical skills and soft skills play a crucial role in talent development and recruitment.

5 THE GAP WITH EDUCATIONAL INSTITUTIONS

Comparisons with previous surveys highlighted a lack of basic 3D CG practice (42.3%) and insufficient production time (52.1%) as the primary issues in developing "specialized technical skills." In the 2011 survey, several educators expressed concerns about the inadequate time allocated for 3D CG practice, indicating that fundamental issues were not being addressed. As a possible solution, many suggested that strengthening collaboration with graduates and companies could alleviate these problems.

Regarding the importance of "cooperativeness," 83.1% of companies responded that it holds significance, and 57.7% of educators also stressed the importance of teamwork, indicating a shared perspective on its relevance. In CG education, project-based CG production is a prevailing method for cultivating cooperativeness, but there is a growing expectation for the development of programs that foster cooperativeness in the arts.

Regarding AI education, the results of the international survey revealed that while educational institutions abroad are actively implementing AI education, those in Japan, despite recognizing the necessity of AI education, face challenges in its implementation. This finding suggests a delay in Japan's efforts to incorporate AI education and underscores the urgent need for adaptation to international educational trends (Supplements Table 3).

6 CONCLUSION

This study reveals a shared emphasis by academia and industry on technical and soft skills such as communication and collaboration. This highlights the urgent need for curricula that combine practical 3D CG skills with collaborative skills to meet industry needs. Challenges such as a shortage of specialized instructors, lack of 3D CG practice, and insufficient specialized curricula indicate the necessity for better collaboration between educational institutions and the industry.

In Japan's 3D CG education, educational institutions need to strengthen collaboration with graduates and companies and introduce projects to meet the specific demands of individual companies. This will enable students to acquire a diverse range of practical skills and understand the needs of the industry. Furthermore, it is essential to introduce new programs that emphasize creativity and communication skills. A notable limitation of this study is the small sample size of survey responses from Japan compared to other countries, potentially leading to reduced statistical power and limited generalizability. Future research should conduct more extensive international surveys with a balanced dataset to provide a more representative perspective on the challenges and potential solutions for Japan's 3D CG education.

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